Data Centers, PJM Capacity Costs, and Residential Electricity Prices

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Abstract

This paper asks whether the recent PJM capacity market shock linked to data center load growth has reached residential electricity prices. The event is the PJM Base Residual Auction for the 2025 to 2026 delivery year. The system price rose from \$28.92 per MW day in the prior delivery year to \$269.92 per MW day, while the Dominion zone cleared at \$444.26 per MW day. I build a monthly utility service territory panel from EIA Form 861M, HIFLD electric retail service polygons, OpenStreetMap data center locations, PJM auction reports, NOAA degree days, and Henry Hub gas prices. The main design compares PJM utilities with non PJM utilities before and after June 2025, and then tests whether the effect is larger for PJM utilities with high data center exposure measured at the 2024 stock. The broad PJM $\times$ post coefficient is 3.2 percentage points. The full panel triple difference for high data center PJM utilities is 12.4 percentage points and remains 10.8 percentage points after adding heating and cooling degree day controls. The historical annual panel from 2010 to 2024 shows no detectable relation between data center counts and residential price growth within state and year cells. The combined evidence is consistent with a timing based mechanism: data center load did not predict retail price growth in the historical period, but the PJM capacity auction created a dated wholesale shock that began passing through to households in 2025.

Keywords: data centers, PJM, residential electricity prices, capacity markets, difference in differences, utility service territory, EIA Form 861M

1 Question and Contribution

Electric utilities and grid operators now face a concrete version of a question that had been mostly speculative before the current artificial intelligence buildout. Data centers require large amounts of dependable power, their load can arrive in geographically concentrated clusters, and their demand is often added faster than generation and transmission supply can adjust. The policy question is whether the resulting wholesale costs are paid mainly by the data centers themselves or spread to households through ordinary retail rate mechanisms.

PJM gives the question a clean test. PJM runs a regional capacity market in which load serving entities buy resource adequacy for future delivery years. The auction for the 2025 to 2026 delivery year cleared in 2024 and applied to June 2025 through May 2026. In that auction, the PJM system price moved from \$28.92 to \$269.92 per MW day. Dominion, the zone that contains the largest Northern Virginia data center cluster, cleared at \$444.26. PJM and its market monitor described data center load growth, combined with supply tightness, as a central reason for the price jump. That makes June 2025 a dated shock to capacity costs rather than a vague trend in electricity demand.

The unit of analysis is the utility by state service territory. This matters because residential prices are not set at the county level and are not observed at the individual data center level. They are reported by utility and state in EIA Form 861M, and they are recovered through state retail regulation, default service procurement, capacity riders, fuel riders, and base rate cases. A county design would not match the billing geography. A state design would mix utilities with very different wholesale exposure. The utility by state panel is the closest feasible unit to the pricing decision.

The paper contributes three things. First, it creates a service territory panel that links residential prices, PJM zone membership, capacity prices, weather, and data center exposure. Second, it separates the historical location selection question from the 2025 pass through question. Data centers tend to locate where power is cheap, so a simple cross section of prices and data center counts is not causal. The historical annual panel is therefore used as a baseline, not as the main treatment design. Third, it uses the PJM delivery year as the event in a triple difference design. The main coefficient asks whether high data center PJM utilities rose after June 2025 by more than other PJM utilities, non PJM high data center utilities, and non PJM utilities with little or no data center exposure.

2 Related Literature

This paper sits between two strands of work. The first studies the scale of data center electricity demand. The International Energy Agency projects that United States data center demand could roughly double by 2026 (International Energy Agency, 2024). Work on artificial intelligence specific load reaches similar conclusions, with estimates that generative model scaling could add tens of TWh of annual demand later in the decade (Ren and Wierman, 2024). Earlier accounts document how data center energy use evolved as efficiency gains offset rising compute (Masanet et al., 2020; Shehabi et al., 2016). These studies establish that the demand shock is large, but they do not trace it into household electricity prices. Interconnection queues in PJM, concentrated in the Dominion zone, now hold large volumes of proposed data center load, on the order of tens of gigawatts in the Dominion zone alone, which the market monitor links directly to the recent capacity price increase (Monitoring Analytics, 2024; PJM Interconnection, 2024).

The second strand studies how wholesale costs reach retail customers. Research on restructured electricity markets shows that regulated utilities recover costs through rate cases that move slowly, while competitive default service can pass wholesale costs through more quickly (Borenstein and Bushnell, 2015; Cicala, 2022). This work explains why a single dated wholesale shock can show up in residential prices at different speeds across utilities. It also motivates a design that allows for both a broad regional response and a slower, more uneven utility level response.

The gap this paper fills is the connection between the two strands. No prior work has measured the residential pass through of this particular PJM capacity shock at the geography where retail prices are set. The paper uses a difference in differences design built around the PJM delivery calendar, in the tradition of applied work that uses local price outcomes to recover the effect of a dated market event (Barron et al., 2021).

3 Institutional Setting

PJM is a regional transmission organization that coordinates wholesale energy, capacity, and transmission planning across all or parts of the Mid Atlantic, Midwest, and South. Its capacity market procures resource adequacy through the Base Residual Auction. Delivery years run from June through May. A capacity price is quoted in dollars per MW day and is paid for committed capacity during the delivery year. Retail customers do not pay that price directly, but utilities and default service suppliers recover capacity costs through retail tariffs.

The 2025 to 2026 auction is the relevant event. The RTO clearing price rose from \$28.92 to \$269.92 per MW day. Dominion cleared at \$444.26, BGE at \$466.35, and most other zones at the RTO price. The jump reflects both sides of the capacity market. Demand rose because expected load increased, with data centers central in Northern Virginia and other PJM zones. Supply tightened because retirements, accreditation changes, and limited new entry reduced reserve margins. This institutional setting supports a research design in which the timing of the cost shock is fixed by PJM rules and the magnitude varies across zones and utilities.

The mechanism runs in a clear sequence. Data center load raises expected peak demand in a zone. Higher expected demand raises the capacity that load serving entities must buy, which lifts the auction clearing price. The higher clearing price raises default service procurement and capacity costs for utilities. Those costs are recovered through rate cases, capacity riders, and default supply charges. Residential retail prices then rise, often with a lag that depends on each state's recovery rules. The empirical design tests whether this channel shows up in residential prices.

Retail pass through is slower and less uniform than wholesale clearing. Some utilities use adjustment charges that can change within a month. Others recover costs through rate cases that take longer. Competitive default service states use procurement rules that can embed capacity costs in default offers. This variation implies that the residential effect need not appear all at once or in the same way for every utility. It also implies that a broad regional effect and a local high exposure effect can coexist.

4 Data Construction

4.1 Residential Prices

The monthly outcome comes from EIA Form 861M (U.S. Energy Information Administration, 2024). For each utility, state, and month, EIA reports residential revenue in thousand dollars, residential sales in MWh, and residential customer counts. The primary outcome is residential price in cents per kWh:

$$\text{Price}_{it} = \frac{100 \times \text{Residential revenue in thousand dollars}_{it}}{\text{Residential sales in MWh}_{it}}. \tag{1}$$

The factor of 100 is important. EIA reports revenue in thousand dollars. Multiplying by 1000 converts to dollars. Dividing by MWh gives dollars per MWh. Dividing by 10 converts dollars per MWh to cents per kWh. The net multiplier is therefore 100. The paper estimates regressions in log residential price. I also calculate the average monthly residential bill as:

$$\text{Bill}_{it} = \frac{1000 \times \text{Residential revenue in thousand dollars}_{it}}{\text{Residential customers}_{it}}. \tag{2}$$

The monthly panel covers January 2023 through March 2026. EIA marks 2023 and 2024 as final and 2025 and 2026 as preliminary. Preliminary status is unavoidable because the post event period begins in June 2025. The analysis keeps utility by state observations with positive residential revenue, positive residential sales, and at least 1000 residential customers for the regression sample. Across the four years the panel holds 20392 utility state month observations covering 563 utility state units across all 50 states and the District of Columbia over 39 months. Of these units, 33 fall inside PJM and form the treated group, split into three high data center units, twelve units with some exposure, and eighteen units with no matched data centers.

Table 1: Coverage of the EIA Form 861M monthly utility service area panel, 2023 to 2026.

Year	Status	Months	Utility states	Observations
2023	Final	Jan to Dec	520	6,068
2024	Final	Jan to Dec	547	6,434
2025	Preliminary	Jan to Dec	549	6,373
2026	Preliminary	Jan to Mar	517	1,517

4.2 Data Center Exposure

Data center locations come from OpenStreetMap records processed into a geocoded facility file. The service territory assignment uses HIFLD electric retail service polygons. Each data center point is matched to the electric service polygon containing its coordinates. The match is then linked to the EIA utility number and state. The assignment summary shows 1550 data center points in the raw file, 1549 with a service polygon match, and 1158 usable matches to EIA state utility pairs in the analysis states. The historical annual panel covers 885 utility state units from 2010 through 2024.

The main exposure variable is the cumulative number of data centers in the utility service territory, measured at the 2024 stock and then held fixed across the monthly panel. Fixing the stock at its 2024 level keeps the treatment from responding to the post June 2025 retail price changes that the design measures, since exposure is set before the event window. I define high data center exposure as 10 or more cumulative data centers, some exposure as 1 to 9, and zero exposure as no matched data centers. The three high exposure PJM utilities are Virginia Electric and Power in Dominion territory with 219 centers, Commonwealth Edison with 46, and Public Service Electric and Gas with 42. Figure 1 shows the cumulative counts across PJM utility service territories.

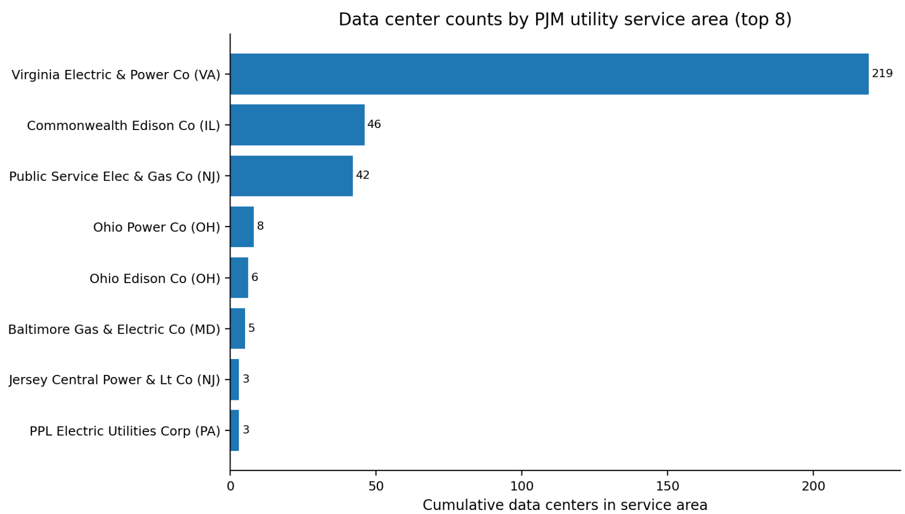

Figure 1: Cumulative data center counts by PJM utility service territory, measured at the 2024 stock. Virginia Electric and Power in the Dominion zone has by far the largest count.

The exposure measure is a count, not MW of data center load. A count is imperfect because a small colocation facility and a large hyperscale campus enter equally. The paper deals with this limitation in two ways. First, the binary high exposure indicator is dominated by the same territories that dominate known data center load, especially Dominion in Northern Virginia. Second, the robustness section uses data centers per 100,000 residential customers as a continuous exposure measure.

Table 2: Descriptive statistics by PJM membership and data center exposure in December 2024.

PJM	DC exposure	N	Mean price (cents/kWh)	Mean customers	Mean DCs
PJM	High (10+ DCs)	3	16.66	2,435,372	102.3
PJM	Some (1 to 9)	12	15.71	534,282	2.9
PJM	Zero	18	16.48	302,005	0.0
Non PJM	High (10+ DCs)	10	13.28	987,506	24.0
Non PJM	Some (1 to 9)	23	13.64	620,654	3.2
Non PJM	Zero	472	21.08	117,867	0.0

4.3 PJM Membership, Capacity Prices, and Controls

PJM membership is assigned with a manually built crosswalk from EIA utility state observations to PJM transmission zones. Utilities outside PJM remain non PJM even when they are located in states with some PJM activity. PJM capacity prices are taken from official Base Residual Auction reports from delivery year 2020 to 2021 through delivery year 2026 to 2027. Each delivery year is expanded to months using the PJM June to May delivery calendar. For example, the 2025 to 2026 auction price applies to June 2025 through May 2026.

Weather controls are state month heating degree days and cooling degree days from NOAA files. Henry Hub monthly gas prices are included in the processed panel as a national fuel cost proxy. The main regressions use month fixed effects, so Henry Hub is absorbed. Weather controls remain useful because they vary across state and month.

4.4 Reproducibility and Folder Audit

The project folder contains a complete analysis path rather than only a final manuscript. The raw data folder stores the source EIA Form 861 annual files, EIA Form 861M monthly files, PJM auction reports, NOAA heating and cooling degree day files, Henry Hub data, HIFLD service territory geometry, and the data center location files. The processed data folder stores the intermediate panels that are used by the regressions. The code folder contains the build scripts, the regression scripts, the machine learning robustness script, and the figure and table script. The figures folder contains the paper ready images and LaTeX tables. The paper folder contains the source file and the compiled PDF.

The core pipeline is organized in four steps. First, the annual EIA Form 861 data are parsed and linked to service territory polygons so that data center exposure can be measured at the same utility state level as retail prices. Second, the EIA Form 861M monthly files are cleaned into a 2023 to 2026 panel and merged with exposure variables. Third, the PJM crosswalk and capacity price files are joined to the monthly utility panel using the delivery year calendar. Fourth, the regression and robustness scripts write numerical outputs that are read by the figure and table script. This structure matters because every number in the paper can be traced to a processed file and a script rather than typed manually into the manuscript.

The folder audit also affects interpretation. The historical annual analysis and the monthly PJM analysis are not two versions of the same regression. They answer different questions with different panels. The historical panel asks whether data center exposure predicted price growth before the PJM event. The monthly panel asks whether a dated PJM capacity shock passed through after June 2025. Keeping those two panels separate prevents the paper from mixing a long run siting question with a short run pass through question.

5 Empirical Strategy

5.1 Historical Baseline

The annual historical panel tests whether data center exposure predicted residential prices before the PJM shock. The specification is:

$$\ln(\text{Price}_{ist}) = \alpha_i + \gamma_{st} + \beta \text{DCExposure}_{ist} + \varepsilon_{ist}. \tag{3}$$

The index i is a utility state unit, s is state, and t is year. The term α_i is a utility state fixed effect and γ_{st} is a state by year fixed effect. The exposure term DCExposure_{ist} is the cumulative data center count in the territory through year t, so it varies within a unit as new facilities open. Identification comes from this within unit growth in the cumulative count, after removing factors common to all utilities in the same state and year. This is a demanding historical baseline because it controls for state policy, fuel mix, and macro conditions that are common within a state year. It is not the main causal design. Its role is to show whether the pre shock data already contained a positive price relation.

5.2 Main Triple Difference

The headline monthly design is:

$$\begin{aligned}\ln(\text{Price}_{it}) = \alpha_i + \gamma_t + \beta_1(\text{PJM}_i \times \text{Post}_t) + \beta_2(\text{HighDC}_i \times \text{Post}_t) \\ + \beta_3(\text{PJM}_i \times \text{HighDC}_i \times \text{Post}_t) + \varepsilon_{it}.\end{aligned} \tag{4}$$

The post indicator equals one from June 2025 onward. The utility fixed effect removes permanent differences across utilities. The year month fixed effect removes shocks common to all utilities in a month, including national fuel prices and national seasonality. Standard errors are clustered by utility state. The regressions are unweighted ordinary least squares, so each utility state cluster enters equally regardless of customer count. The descriptive figures and the zone level tables are customer weighted instead. This difference in weighting is one reason the regression coefficients differ in size from the customer weighted quarterly gaps, since the regressions give a small utility the same weight as a large one while the descriptive series are dominated by large utilities.

The coefficients have direct interpretations. β_1 is the broad PJM uplift after the capacity shock. β_2 is the high data center change outside PJM, which functions as a placebo for the PJM mechanism. β_3 is the central estimate. It measures the extra post June 2025 change for high data center utilities inside PJM after netting out the broad PJM change and the high data center change outside PJM.

Because the outcome is logged, the coefficients are close to percentage point changes for the magnitudes in this paper. A coefficient of 0.032 is read as about 3.2 percent higher residential prices after the event. A coefficient of 0.124 is read as about 12.4 percent higher prices for high exposure PJM utilities relative to the relevant comparison changes. The exact exponential conversion gives nearly the same economic reading at these magnitudes, so the tables report coefficients in percentage points for clarity.

The full triple difference can also be described in plain group means. Start with the post event price change for high exposure PJM utilities. Subtract the post event change for lower exposure PJM utilities. This removes the broad PJM shock that affects the region. Then subtract the same high exposure versus lower exposure difference outside PJM. This removes any national pattern in data center territories that is not specific to PJM. The remaining term is the estimate of the PJM specific high data center differential after June 2025.

5.3 Identification

The design rests on four assumptions. First, treatment status is predetermined. PJM membership and the 2024 data center exposure stock are fixed before June 2025. Second, the timing of the shock is institutional. June 2025 is set by the PJM delivery calendar, not by utility pricing decisions. Third, the relevant groups follow similar pre event trends after accounting for fixed effects. The raw price indexes are consistent with this condition through spring 2025. Fourth, no other shock occurs at the same time that both targets PJM and differentially targets high data center service territories. Weather is the main concern, which is why the paper reports a weather controlled version on the NOAA coverage sample.

The most important threat is not that data centers choose cheap locations. That threat is real for cross sectional models, but it is mostly absorbed here by utility fixed effects and by holding exposure fixed at the predetermined 2024 stock. The larger threat is that high exposure PJM utilities might have had different scheduled rate cases or riders in 2025 for reasons unrelated to data center load. The design cannot observe every tariff mechanism directly. It responds by using non PJM high exposure utilities as a comparison, by checking weather controls, by comparing within PJM groups, and by reporting the short post event timing in the event figure. These checks do not make the design experimental, but they make the identifying claim narrow and testable.

6 Results

6.1 The PJM Shock

Figure 2 shows the jump in PJM capacity prices. The visual break is clear at the 2025 to 2026 delivery year. The RTO price increases sharply and Dominion separates even further from the system price.

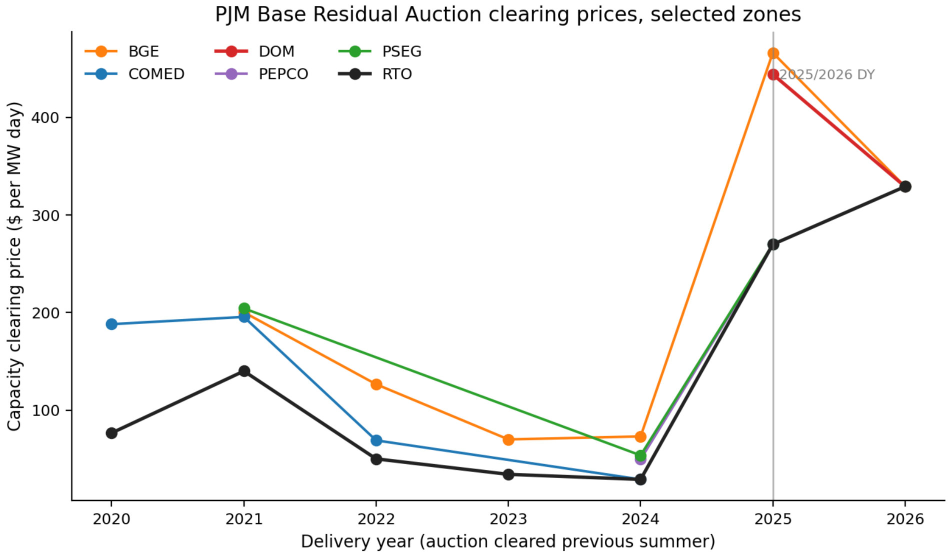

Figure 2: PJM Base Residual Auction clearing prices for selected zones and the RTO system price. The 2025 to 2026 delivery year is the dated capacity cost shock used in the design.

Figure 3 plots the customer weighted residential price index for PJM and non PJM utilities. The two series move together through most of the pre event period. In June 2025, the PJM index jumps sharply while the non PJM index does not. By March 2026, the PJM index is roughly 18 points above the non PJM index.

Figure 3: Customer weighted residential price index with January 2023 equal to 100. PJM prices accelerate after the June 2025 delivery year begins. EIA data for 2025 and 2026 are preliminary.

6.2 Descriptive Zone and Exposure Patterns

The zone level pattern is broad. Table 3 shows customer weighted residential price changes from Q1 2025 to Q1 2026. Dominion rises 19.0 percent, PSEG 17.4 percent, BGE 14.9 percent, PECO 13.5 percent, and PPL 13.7 percent. COMED rises less, at 6.9 percent. The result is not only a Dominion story. Many PJM zones show double digit residential price growth.

Table 3: Customer weighted residential price change by PJM zone, Q1 2026 versus Q1 2025. Data for 2025 and 2026 are preliminary.

Zone	Utilities	Customers (mn)	Q1 2025 cents/kWh	Q1 2026 cents/kWh	% change
DAY	1	0.13	14.86	18.93	+27.4%
AE	1	0.47	22.66	27.98	+23.5%
DOM	4	2.77	13.91	16.56	+19.0%
PEPCO	3	0.96	18.73	22.05	+17.7%
PSEG	1	1.97	21.27	24.97	+17.4%
JCPL	1	0.95	16.00	18.63	+16.4%
BGE	1	1.08	19.12	21.96	+14.9%
DLCO	1	0.44	22.06	25.21	+14.3%
PPL	1	0.81	17.16	19.51	+13.7%
METED	1	1.46	17.83	20.24	+13.6%
PECO	1	1.21	16.45	18.68	+13.5%
ATSI	3	0.47	14.94	16.67	+11.6%
DEOK	2	0.41	14.41	15.75	+9.3%
COMED	1	2.96	16.51	17.64	+6.9%
DP&L	2	0.45	18.70	19.79	+5.8%
AP	3	0.72	14.25	14.88	+4.4%
AEP	6	2.01	16.83	17.32	+2.9%

Figure 4 groups the same Q1 change by PJM membership and data center exposure. PJM high exposure utilities rise 14.5 percent on a customer weighted basis. PJM some exposure and zero exposure utilities also rise substantially, at 11.2 percent and 12.1 percent. Non PJM groups rise less. This descriptive evidence points to a regional PJM effect plus a smaller high exposure component.

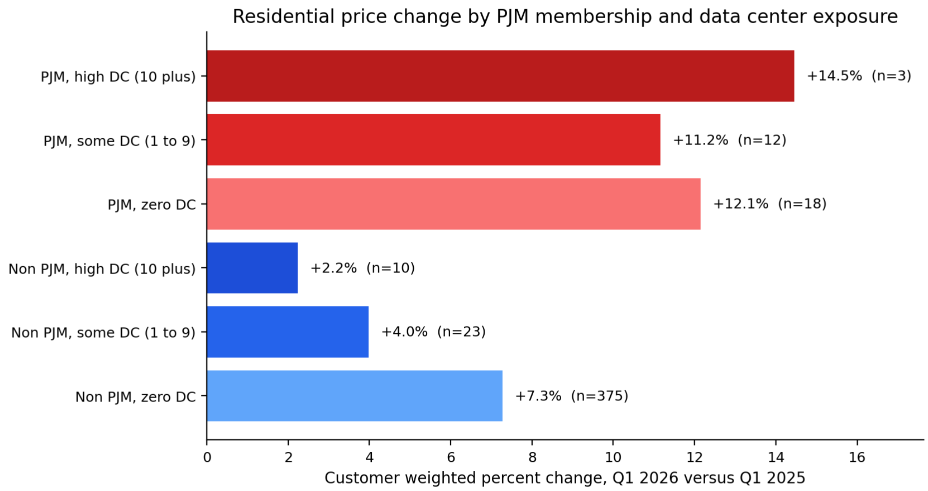

Figure 4: Customer weighted residential price change from Q1 2025 to Q1 2026 by PJM membership and data center exposure group.

The within PJM parallel trends plot in Figure 5 rebases each group to May 2025. The high exposure PJM group and the low or zero exposure PJM group are close before the event. Both jump in June 2025, but the high exposure group moves more. Non PJM utilities do not show the same immediate increase.

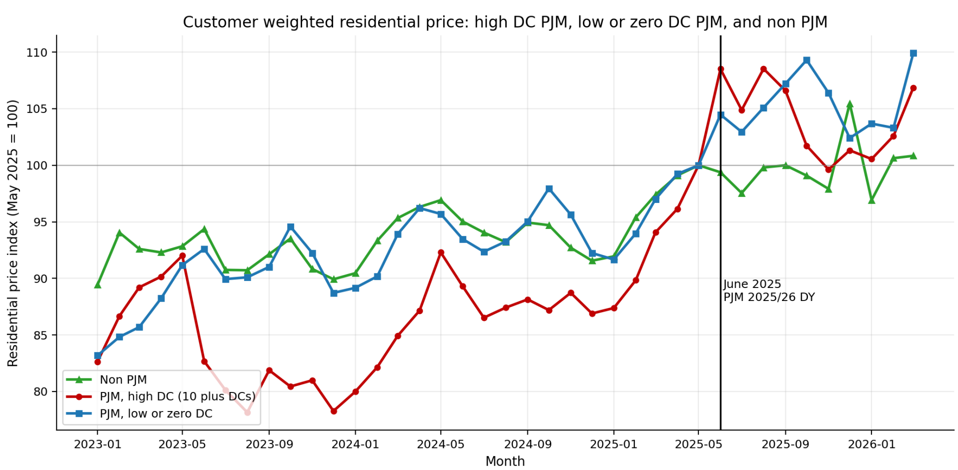

Figure 5: Customer weighted residential price index with May 2025 equal to 100. The graph separates high data center PJM utilities, low or zero data center PJM utilities, and non PJM utilities.

6.3 Main Regression Results

Table 4 reports the core estimates. The historical annual coefficient on data centers per 100,000 customers is 0.02 percentage points below zero and statistically indistinguishable from zero. This supports the location selection concern: before the PJM shock, data center exposure did not predict faster residential price growth within state and year comparisons.

Table 4: Main regression results. Outcome is log residential price. Standard errors are clustered by utility state. Coefficients are reported in percentage points with SE in parentheses. All monthly specifications include utility state and month fixed effects. The historical specification uses utility state and state by year fixed effects.

Specification	Sample	Estimate	p value	N
Historical annual: DCs per 100k cust.	2010 to 2024 utility state	-0.02 (0.02)	$p = 0.331$	9,136
Monthly pass through: log capacity	PJM utilities 2023 to 2026	$+1.16$ (3.92)	$p = 0.768$	358
Within PJM: High DC × post	PJM utilities 2023 to 2026	$+8.05$ (3.66)	$p = 0.028$	1,275
Within PJM: Any DC × post	PJM utilities 2023 to 2026	$+1.69$ (2.88)	$p = 0.558$	1,275
Broad uplift: PJM × post	Full panel 2023 to 2026	$+3.22$ (1.52)	$p = 0.034$	17,457
Triple DID: PJM × high DC × post	Full panel 2023 to 2026	$+12.41$ (5.01)	$p = 0.013$	17,457

The monthly triple difference gives the main result. The broad PJM by post coefficient is 3.22 percentage points with a p value of 0.034. The placebo coefficient β_2, which is the high data center change outside PJM, is 4.33 percentage points below zero with a p value of 0.219, so there is no separate national high exposure effect once PJM is accounted for. The PJM by high data center by post coefficient is 12.41 percentage points with a p value of 0.013. The within PJM comparison, which drops non PJM controls and compares high exposure PJM utilities with other PJM utilities, gives 8.05 percentage points with a p value of 0.028. These estimates are larger than the simple customer weighted Q1 gap because the regression uses within utility changes and equal cluster treatment rather than only customer weighted aggregates.

The monthly pass through regression using the log capacity price inside PJM is not significant. That does not contradict the event design. Once month fixed effects absorb the system wide jump, little independent capacity price variation remains within PJM zones. The pass through question is better identified by the June 2025 event and exposure structure than by a continuous zone price regression over a short window.

The size of the estimates is economically meaningful. A 3.2 percent increase on a 17 cent per kWh residential price is about 0.54 cents per kWh. For a household using 900 kWh in a month, that is roughly \$4.90 per month. The high exposure triple difference is larger. A

12.4 percent increase on the same base price is about 2.1 cents per kWh, or roughly $19 per month for 900 kWh. These are simple translations, not separate estimates, but they show why capacity market pass through matters for household bills even when the wholesale price is quoted in MW day units rather than cents per kWh.

6.4 Weather Controls

Weather is the most plausible competing explanation because winter demand differs by region. Table 5 restricts the sample to states with NOAA heating and cooling degree day coverage and adds HDD and CDD controls. The broad PJM coefficient falls to 1.33 percentage points and is not significant. The triple difference remains large at 10.80 percentage points with a p value of 0.005. This means the broad PJM versus non PJM comparison is partly sensitive to weather, while the high exposure PJM differential is not.

Table 5: Triple difference design with weather controls. Outcome is log residential price. Sample is utility states with NOAA HDD and CDD coverage. Unit and month fixed effects. SEs are clustered by utility state. Coefficients are in percentage points. HDD and CDD are reported per 1,000 degree days.

Treatment	Estimate	SE	p value	N
PJM × post shock	+1.33%	2.07	$p = 0.521$	4,215
High DC × post shock	−2.45%	1.79	$p = 0.172$	4,215
PJM × high DC × post shock	+10.80%	3.80	$p = 0.005$	4,215
HDD (thousands)	+29.00%	14.34	$p = 0.043$	4,215
CDD (thousands)	−38.64%	22.72	$p = 0.089$	4,215

6.5 Exposure and Sample Robustness

The high exposure cutoff is not the only way to code treatment. Table 6 reports four cutoffs and a continuous exposure model. The triple difference is positive at each threshold. At 5 data centers, the estimate is 7.65 percentage points. At 10, the headline cutoff, it is 11.97 percentage points in the threshold sensitivity sample. At 20, it is 11.84 percentage points. At 50, only Dominion remains in the high exposure PJM group, so that estimate is reported but should not be treated as a broad population coefficient.

Table 6: Triple difference robustness to the high DC threshold and to continuous exposure. Outcome is log residential price. Specification follows Equation 4 with utility state and month fixed effects. SEs are clustered by utility state. The triple coefficient is either PJM × HighDC × post or PJM × DC per 100k × post. Sample has $N = 6,377$ utility state month observations.

Specification	High DC PJM units	Triple coef.	SE	p value	PJM×post
Panel A: Binary high DC threshold sensitivity					
Threshold ≥ 5 DCs	6	+7.65%	4.18	0.067	+3.49%
Threshold ≥ 10 DCs (headline)	3	+11.97%	5.08	0.018	+3.66%
Threshold ≥ 20 DCs	3	+11.84%	6.31	0.061	+3.81%
Threshold ≥ 50 DCs	1	+7.35%	1.72	<0.001	+4.71%
Panel B: Continuous exposure (DCs per 100k residential customers)					
Per additional DC per 100k cust.		+1.02 pp	0.60	0.092	+4.13%
Implied: Dominion VA (≈9.2 DC/100k)		+9.35 pp	5.54		
Implied: ComEd IL (≈1.6 DC/100k)		+1.60 pp	0.95		
Implied: PSE&G NJ (≈2.1 DC/100k)		+2.17 pp	1.29		

The smaller sample in this table, $N = 6{,}377$ against 17,457 in the headline triple difference, reflects a coding choice rather than a different population. The threshold and continuous specifications condition on a non-missing cumulative data center count, so they retain only utility state units that were successfully linked to the data center assignment file. The headline regression instead treats unlinked units as zero exposure and keeps the full panel, which is why its sample is larger.

The continuous version estimates 1.02 percentage points per additional data center per 100,000 customers for PJM utilities after June 2025. The implied effect is 9.35 percentage points for Dominion, 1.60 for ComEd, and 2.17 for PSEG. This shows that the binary high exposure estimate is not uniform across the three high exposure utilities. It is concentrated most strongly in Dominion, which has the highest data center density.

Table 7 restricts the sample to service territories that do not cross state lines. This was included because some service territory polygons span more than one state. The limitation is severe: only two PJM utility state observations remain and none are high exposure PJM utilities. The broad PJM coefficient remains positive but imprecise. The triple difference cannot be identified in this strict sample. The result is useful mainly as a transparency check, not as a substitute for the headline design.

Table 7: Robustness restricted to single state utility service areas. The sample is utility states whose HIFLD service territory polygon does not cross state lines. It contains 48 utility states, 2 in PJM, and 0 high DC PJM units. Unit and month fixed effects are included. SEs are clustered by utility state. The triple DID interaction is not identified on this subsample because no high DC utility has a single state service territory polygon.

Treatment	Estimate (pp)	SE (pp)	p value	N
PJM × post (broad uplift)	+5.72	4.82	$p = 0.235$	1,872
High DC × post outside PJM	−1.21	0.61	$p = 0.048$	1,872
PJM × high DC × post triple DID	Not identified in this subsample			1,872

6.6 Machine Learning Robustness

The project also includes three machine learning checks. They are not used as black box causal estimators. Each one targets a specific weakness in the main design.

First, double selection Lasso on the historical annual panel chooses controls from residential sales, customer counts, commercial customers, industrial sales, and log transformations (Belloni et al., 2014). The selected control set produces a data center coefficient of 0.010 percentage points with a p value of 0.487. This confirms the historical null result after data driven control selection.

Second, a random forest is fit to residual monthly residential prices after removing utility and month fixed effects, in the spirit of flexible machine learning checks on the role of treatment variables (Athey and Wager, 2019; Chernozhukov et al., 2018). The random forest does not prove causality, but it shows that the variables central to the research design carry predictive signal in the processed panel.

Third, a wild cluster bootstrap resamples residuals at the utility state cluster with 999 Rademacher replications (Cameron et al., 2008). The bootstrap p value for the triple difference is 0.015, close to the analytical cluster p value from the same routine.[1] This addresses the concern that the high exposure group has few clusters.

[1] The bootstrap routine re-estimates the triple difference inside its own resampling pipeline, which yields a point estimate of 12.21 percentage points with an analytical cluster p value of 0.015, marginally different from the 12.41 percentage points and p value of 0.013 reported in Table 4; the broad PJM uplift is 3.43

The reason for including these checks is practical. A conventional fixed effect design is transparent, but it leaves three common objections. One objection is that the control set in the historical panel could have been chosen after seeing the result. Lasso responds to that by selecting controls through a rule. A second objection is that the core variables might not matter once flexible prediction is allowed. The random forest responds by ranking the variables in a nonlinear prediction exercise after fixed effects are removed. A third objection is that the number of high exposure clusters is small. The wild bootstrap responds by using a cluster based resampling procedure designed for that inference problem. None of the checks replaces the main design, but all three point in the same direction as the main interpretation.

Table 8: Machine learning robustness specifications. Panel A reports Lasso post selection on the historical 2010 to 2024 utility state annual panel. Panel B reports a wild cluster bootstrap of the triple difference coefficients with 999 Rademacher replications at the utility state cluster. Bootstrap p values are reported alongside analytical cluster robust p values.

Panel A: Lasso post selection on historical 2010 to 2024 panel

Specification	Estimate (pp)	SE (pp)	p	N	Selected / Cand.
Double selection Lasso	+0.010	0.014	$p = 0.487$	9,136	7 / 7

Panel B: Wild cluster bootstrap on triple difference with 999 Rademacher replications

Coefficient	Estimate (pp)	SE (pp)	Analytical p	Bootstrap p	N
PJM × post (broad uplift)	+3.43	1.51	0.023	0.019	17,457
High DC × post outside PJM	−4.13	3.48	0.236	0.248	17,457
PJM × high DC × post triple DID	+12.21	4.98	0.015	0.015	17,457

7 Interpretation

The estimates support a two part interpretation. The first part is regional pass through. PJM residential utilities as a group raised prices after the 2025 to 2026 capacity delivery year began. This is visible in raw customer weighted indexes and in the broad PJM by post regression coefficient. The second part is high exposure pass through. Within PJM, the utilities with the largest data center footprint rose by more than other PJM utilities after the shock.

rather than 3.22 percentage points across the two routines. Both estimates are significant at the five percent level. The small gap reflects the separate estimation pipeline and does not affect any conclusion.

This pattern fits the institution. Capacity costs are regional because PJM clears a regional market and load serving entities buy resource adequacy through that market. At the same time, localized constraints and state retail recovery rules can make high exposure territories move more. Dominion faces the clearest local data center load pressure. PSEG shows a sharp retail movement through its adjustment mechanisms. ComEd has high facility counts but a smaller Q1 2026 price increase, which is why the continuous exposure model shows that the binary result is driven more by Dominion than by a uniform effect across all three high exposure utilities.

The historical null is not a weakness of the paper. It is part of the argument. If data centers selected cheap power locations before the capacity shock, then pre 2025 cross sectional comparisons should not show data centers raising prices. That is exactly what the annual panel finds. The change after June 2025 is different because the PJM auction creates a dated wholesale cost shock that can be observed in residential prices.

8 Limitations

The main limitation is exposure measurement. Counts of data center facilities are not the same as MW of load. A better treatment variable would measure committed or realized data center demand by utility territory. That would require linking campuses to load estimates from interconnection queues, utility filings, commercial data center sources, or FERC demand records. The current count based measure is defensible for ranking territories, but it understates load differences among large campuses.

The second limitation is the short post period. The panel ends in March 2026, only ten months after the delivery year begins. Some capacity costs pass through quickly through riders. Others may enter retail prices through future base rate cases. The estimates should be read as early pass through, not full long run incidence.

The third limitation is the small high exposure PJM group. The headline high exposure utilities are Dominion, ComEd, and PSEG. Clustered standard errors and the wild bootstrap help, but the external validity of the high exposure coefficient still rests on few units. The threshold and continuous exposure checks are included to show that the sign and scale are not artifacts of one arbitrary cutoff.

The fourth limitation is preliminary data. EIA can revise 2025 and 2026 monthly Form 861M data. Large utilities tend to have more stable reporting than small utilities, and the customer weighted descriptive results are driven by large utilities. Even so, the final annual files should be used to rerun the analysis when they become available.

9 Conclusion

The evidence shows that the PJM capacity shock associated with data center load growth has started to reach residential electricity prices. By Q1 2026, PJM residential prices were rising faster than non PJM prices, and high exposure PJM utilities rose more than other PJM utilities after controlling for utility and month fixed effects. The main triple difference is 12.4 percentage points and remains 10.8 percentage points with weather controls. The historical annual panel shows no comparable effect before 2025, which supports the view that the key change is the dated PJM capacity shock rather than a simple correlation between data centers and high prices.

For regulators, the result shifts the question from whether data center load can create household costs to how those costs should be allocated. If new capacity costs are recovered through broad residential rates, households pay part of the cost of serving concentrated commercial load. If costs are assigned more directly to the customers that drive the new load, the incidence changes. This paper does not answer the normative allocation question. It suggests that the allocation question is already active in residential bills.

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